

NETRA

Network observability and emergency control without CNI lock-in.

A buyer brochure for enterprises evaluating Netra — how every major capability works, what the P0–P5 observe Surfaces add, how incidents move from signal to containment, and how Netra sits next to PacketWolf in the Zyvor suite.

OBSERVE · DIAGNOSE · CONTAIN · FAIL OPEN · Standalone eBPF for Linux/Kubernetes · Apache-2.0 · Optional Cilium/Hubble enrichment

When enterprises buy Netra

Modern clusters accumulate network tools faster than operators can trust them. CNI dashboards explain policy verdicts. Host stacks still drop packets in softnet, qdisc, and the NIC. Emergency denies that never expire become the next outage. Buyers also ask for encrypted-traffic intel, unsanctioned SaaS visibility, and GenAI destination awareness — without decrypt or payload harvest. Netra is built for that gap.

Buyer problems Netra addresses

- CNI-only visibility leaves node-level loss unexplained.
- Changing or mixing CNIs should not erase observability and emergency control.
- Encrypted traffic must be understandable without SSL inspection.
- Incident kill-switches must auto-revert — permanent silent deny is unacceptable.
- Evidence (capture, alerts, SIEM, AI) must not collect payloads, argv, or secrets.

What Netra is — in one sentence

Netra is standalone eBPF network observability and leased emergency containment for Linux/Kubernetes: observe-first, CNI-independent by default, with optional Cilium/Hubble enrichment.

Architecture — how it works

Netra has two workloads: a controller (netrad) that serves the API/UI and holds desired state, and a privileged node agent that owns eBPF programs and maps under `/sys/fs/bpf/netra`.

01 Workloads — Pods · VMs · host processes

02 eBPF hooks — cgroup skb · connect/sendmsg · sockops · optional TCX/XDP

03 Netra maps — `/sys/fs/bpf/netra` — agent-owned; never Cilium maps

04 Agent — Reports · capture stream · lease sync · JA3 samples

05 netrad — API · Surfaces UI · HA leader · SIEM/notify

The agent never pins over Cilium-owned maps. Cilium and Hubble are read as optional data sources.

Surfaces — P0–P5 observe boards

The console Surfaces page (Diagnostics + Security) boards every metadata fit surface. All boards are observe-only; apply still requires an enforce lease + risk confirm and fails open.

Encrypted

- Always-on JA3/JA4 from standalone `bpf/netra_tlsfp.c` (plus capture frames).
- DoT (port 853) and DoH (known hosts via SNI/Host/DNS).
- JA3 risk and ECH / missing-SNI blindness boards.

Fit (SSE/ZT-adjacent, no SWG)

- Shadow SaaS vs NETRA_SANCTIONED_HOSTS allow-list.

- Workload digital-experience scorecard (latency / retrans / DNS fail).
- Destination risk ranking (intel + categories + exposure + volume).
- Zero Trust / microseg / policy-pack / identity drafts — review-only.

Threat & ops

- Live threat-intel feed → hits → optional leased apply.
- AI/MCP SaaS destination catalog + optional leased SNI deny.
- Exfil fan-out heuristics (not DLP), lateral playbooks, category deny drafts.
- App categories, CIS-inspired compliance pack, prevention coverageScore.
- Opt-in volumetric auto-mitigate (SYN/UDP) behind an active lease.

Fleet

- Multi-cluster read-only aggregator (NETRA_FLEET_PEERS).
- Tenant risk scores for partner / MSSP rollups.

Technical catalog: [docs/p0-p5-surfaces.md](#) · Buyers guide: [docs/sales/buyers-guide.md](#)

Observability

Netra records flow counters, direction, hook attribution, TCP/UDP/ICMP visibility, Kubernetes workload identity, and metadata-only L7 signals including JA3/JA4 samples.

Hard boundary: no application payloads. No argv/cmdline. No Secret contents. DNS is ordinary UDP/53 query names. TLS is best-effort ClientHello metadata in a single skb. HTTP is method + Host only.

Path, Drop, Congestion Map & Capture

Path Diagnostics measures connect-establishment latency and transport pressure per workload. Drop Diagnostics combines optional kfree_skb reasons with softnet / interface / qdisc counters. Congestion Map paints every stack stage by severity. Capture is filtered and time-bounded (eBPF or AF_PACKET); opt-in auto-capture writes classic PCAPs on critical signals.

Emergency firewall — leased enforce

Netra's containment layer is deliberately an emergency tool — not a full CNI policy engine, QoS system, L7 proxy, or IDS/IPS.

- Exact IPv4/IPv6 and CIDR deny (ingress/egress), optional SYN-drop mode
- Port, DNS name, TLS SNI, UID, process comm — with allow-exceptions
- PPS/BPS ceilings and per-workload new-TCP connection-rate caps
- Optional XDP early ingress on selected interfaces

Lifecycle: Observe (default) → Plan + confirm → Enforce (leased) → Auto-revert (fail open). Blast-radius preview matches a proposed deny against live counters and applies nothing.

Alerting, export, AI, MCP & ChatOps

Multi-channel notify (webhook, email, Slack, Teams, Twilio, HTTP bridge) with HA leader-only delivery. Pull SIEM export supports JSON, JSONL, CEF, syslog, and OTLP. Ask Netra / digests / draft are read-only — they never flip enforce. netra-mcp mutations stay gated off by default. Slack/Teams ChatOps mirrors UI confirmations.

Zyvor suite placement — Netra & PacketWolf

Netra and PacketWolf are suite counterparts, not a wired pipeline. Neither product is a dependency or API consumer of the other today.

Netra — Mixed / non-Cilium entry; observe + Path/Drop/Congestion + leased emergency deny; owns /sys/fs/bpf/netra only

PacketWolf — Cilium-standardized fleets; observe + AutoPolicy + healer + platform containment

- Same cluster is allowed but not required — Netra maps stay isolated.
- Do not dual-own long-lived deny: PacketWolf/Cilium for durable posture; Netra leases for incident kill-switch.
- There is no Netra ↔ PacketWolf control-plane sync today.

Buying checklist

- Confirm Linux nodes meet cgroup v2 / bpffs / BPF baselines (netra-doctor).
- Start observe-only: flows, Path, Drop, Congestion Map, Surfaces.
- Generate TLS traffic and confirm JA3 on Surfaces / L7; set NETRA_SANCTIONED_HOSTS for shadow SaaS.
- Load a small intel feed; do not apply until a lease drill.
- Exercise capture (and optionally auto-capture) on a non-production signal.
- Pilot a leased deny with blast-radius preview and confirm auto-revert.
- Wire alerting / SIEM / ChatOps; keep AI/MCP read-only until trusted.
- If Cilium is present, decide enrichment vs PacketWolf platform ownership explicitly.

See the network. Contain the incident. Fail open by default.

Talk to sales → zyvor.dev/netra · sales@zyvor.dev

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